Candidate's Assessment Number: _____

TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM TWO INTER ISLAMIC MOCK EXAMINATION

BIOLOGY

033

TIME: 2:30 HOURS **Wednesday, 24th July 2024 a.m.**

Instructions

1. This paper consists of sections A, B and C with a total of ten (10) questions.
2. Answer all questions in each section.
3. All writings must be in blue or black ink except drawings which must be in pencil.
4. Cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your **Examination Number** at the top right corner of every page.

FOR EXAMINERS' USE ONLY		
QUESTION NUMBER	**SCORE**	**EXAMINER'S INITIALS**
1		
2		
6		
7		
8		
9		
10		
TOTAL		
CHECKER'S INITIAL		

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SECTION A (15 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter in the box provided.

(i) Mbumba went to keep his cattle in the forest whereby he observed an organism with cell wall made up of Chitin. As a recent taxonomist, in which taxa would you group such organism?

A. Kingdom plantae
B. Kingdom Fungi
C. Kingdom Protoctista
D. Kingdom Monera

(ii) A Process in which plants and green algae use simple inorganic substances to produce glucose is called _____

A. Hydrolysis
B. Photolysis
C. Photosynthesis
D. Plasmolysis

(iii) A mother refuses to breast feed her baby. What type of immunity will that baby miss?

A. Natural passive Immunity
B. Natural active Immunity
C. Artificial passive immunity
D. Artificial active immunity

(iv) A form one student went to hospital laboratory for the check-up whereby he was diagnosed with the sleeping sickness. Suggest the possible vector of such disease.

A. Tsetse fly
B. Mosquitoes
C. Butterfly
D. Housefly

(v) Assume you have been assigned by school manager to cross-section cutting down the mango stem. Recommend on the arrangement of vascular bundle.

A. Random pattern
B. Scattered pattern
C. Ring pattern
D. Peripheral pattern

(vi) Mr. Tondwa being a street vendor, got an accident that caused bleeding through the cuts when he tried to slash the long grasses around Mr. Ally's house. As Mr. Ally what would you do to help before taking him to the hospital?

A. Lay Mr. Tondwa or his back while legs rise
B. Dress Mr. Tondwa's wound and bandage it firmly
C. Give Mr. Tondwa a lot of fruit juice to restore mineral lost
D. Induce vomiting to Mr. Tondwa

(vii) A term that describes the position that an organism occupies in a food chain due to their feeding habit is called _____

A. Ecosystem
B. Food web
C. Trophic level
D. Food chain

(viii) Euglena is normally mislead as a member of Plantae kingdom which is not true because plants are not locomoting. Based on this statement what feature of euglena will you use to separate them from plants?

A. Flagellum
B. Pseudopodia
C. Cilia
D. Legs

(ix) When form two students poured Iodine solution into the test tube containing a food sample X it turned blue-black. Comment on the colour change.

A. The test tube contains glucose
B. The test tube contains lipid
C. The test tube contains protein
D. The test tube contains starch

(x) Recommend the biological term that explains the substance which poses the threat to the health of living organism as a causing agent of disease and infection.

A. Fragile
B. Biohazard
C. Corrosive
D. Flammable

2. Match the descriptions of diseases and infections in LIST A with the corresponding terms in LIST B by writing the letter of the correct responses beside the item number.

	LIST A	**LIST B**
(i)	Disease that affects a wide geographical area	A. Vaccination
(ii)	The inversion of the body by disease-causing microorganism	B. Infection C. Pandemic
(iii)	A way of inducing resistance against an antigen	D. Disease
(iv)	Condition that affects health of the body and characterized by a certain symptom	E. Endemic F. Pathogen
(v)	A microorganism that can affect health if gets into the body	G. Passive immunity H. Communicable disease

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B					

SECTION B (70 Marks)

Answer **all** questions in this section

3. Recent environmentalist publications and reports condemn on the Serious environmental pollution resulted from poor waste disposal.
   (a) If you could be assigned to educate people on how condemned issue can keep them unsafe, use four points to do so.
   (i) _____
   (ii) _____
   (iii) _____
   (iv) _____

   (b) "Dispose waste properly to avoid environmental pollution." Use three main principles to support this statement.
   (i) _____
   _____
   (ii) _____
   _____
   (iii) _____
   _____

4. While performing surgery on mammal, two blood vessels were cut mistakenly whereby in one of the vessels blood flows out with high pressure while in the other vessels blood flow out with low pressure.
   (a) As a recent medical doctor suggest names of the two blood vessels
   (i) _____
   (ii) _____
   (b) Assess four differences between vessels mentioned in (a) above

(i)		
(ii)		
(iii)		
(iv)		

5. Briefly explain how the following processes are involved in carbon cycle.
   (i) Respiration _____
   _____
   (ii) Combustion _____
   _____
   (iii) Fossilization _____
   _____
   (iv) Photosynthesis _____
   _____
   (v) Decomposition _____
   _____

6. $2cm^3$ of a food sample solution Q changed to purple colouration when $1cm^3$ of sodium hydroxide were added on it followed by 2-3 drops of copper II sulphate solution.
   (a) Identify the food substance present in the solution Q _____

   (b) State any two enzymes responsible for digestion of food substance Q and their respective products.
   (i) _____
   (ii) _____

   (c) Give five roles of food substance contained in the solution Q.
   (i) _____
   (ii) _____
   (iii) _____
   (iv) _____
   (v) _____

7. On the way to Morogoro Chaupele observed road accident that caused injuries to some passengers, when asking himself about what to do to help the victims he found the small box with read-cross on its top in vehicle.
   (a) Suggest and explain the term used to name the box
   _____
   _____

   (b) Explain four (4) items from the box that would be used by Chaupele to assist the victims
   (i) _____
   _____
   (ii) _____
   _____
   (iii) _____
   _____
   (iv) _____
   _____

8. (a) Classifying organisms is among of key areas in studying biology. Use the knowledge of biology classification you have to complete the following table:

	ORGANISM	KINGDOM	PHYLUM
(i)	Common mould		
(ii)	Moss plant		
(iii)	Paramecium		
(iv)	Mushroom		
(v)	Fern plant		
(vi)	Yeast		

8. (b)"Plantae and fungi are not identical groups of organisms." Give two reasons to support this fact.

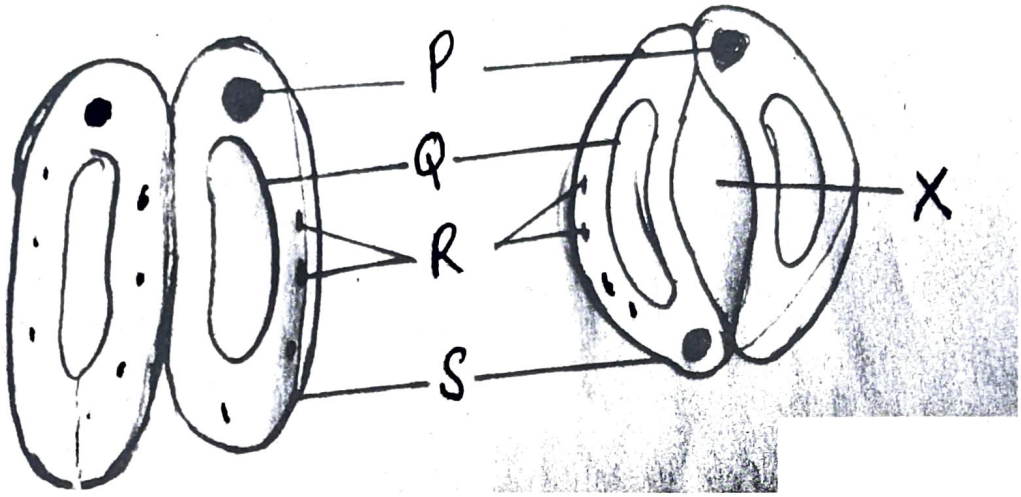
   (i) _____

   _____

   (ii) _____

   _____

9. Observe the figure 01 below and attempt the questions that follow:



(a) Name the structure represented by figure 01 above and its specific role in plant body.

_____

_____

(b) Name the parts labelled P, Q, R and S and state role played by each part.

PART	ROLE
P.	
Q.	
R.	
S.	

(c) How is plant's body benefited from the process carried out by figure 01? (Give three (3) facts)

(i) _____

(ii) _____

(iii) _____

SECTION C (15 Marks)

Answer question **ten (10)**

10. The body of living organism has a means of generating energy either oxygen is available or not. Use six (6) biological facts to explain the difference between these two scenarios.

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MARKING SCHEME

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	C	A	A	C	B	C	A	D	B

01 = @ = 10 Marks

2. Matching items

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B	C	B	A	D	F

01 @ = 05 marks

3. (a) (i) Outbreak of diseases

(ii) Pollute the aquatic environment and death of aquatic animals.

(iii) Waste can block drainage channels and cause flood.

(iv) Waste may degrade the natural beauty of our environment

(v) land pollution

(vi) Air pollution

(vii) Soil Pollution

(viii) water Pollution

(ix) Loss of biodiversity

@ 1 x 4 [04 marks]

(b) (i) Recycling: Changing of waste into usable products, e.g. broken grass can be melted to make new glass bottles.

(ii) Re-using: - Uses of waste in other productive (beneficial) purpose e.g. use of empty margarine container to store sugar.

(iii) Reducing: Act of minimizing the amount of waste produced e.g. Uses of women baskets instead of plastic bags.

@ 2 x 3 = 06 marks

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4. (a) (i) Artery

(ii) Veins **@ 1 x 2 = 02 marks**

Artery		Vein
(i)	Carries oxygenated blood except pulmonary artery.	Carries deoxygenated blood except pulmonary veins
(ii)	Carries blood away from the heart	Carries blood towards the heart
(iii)	Pumps blood in high pressure	Pumps blood in low pressure
(iv)	Have thick muscular wall	Have thin muscular wall
(v)	Have narrow lumen	Have wide lumen
(vi)	Has no Valve	Have valve

(any four points)

@ 2 x 4 = 08 marks

5. (i) Respiration : Oxidation of food to release energy. After breaking down of food oxygen is released as by-product then released to the atmosphere.

(ii) Combustion: Refers to the burning of substance in presence of oxygen. This process contributes carbondioxide that released from burning materials to the atmosphere.

(iii) Fossilization: Remains of plants and animals results into formation of fuels that when burnt release carbondioxide into the atmosphere.

(iv) Photosynthesis: During photosynthesis carbondioxide from atmosphere is absorbed by plants and used to carryout photosynthesis.

(v) Decomposition:- When a plant and animal die thy are decomposed by microorganism to release carbondioxide gases to the atmosphere.

@ 02 x 05 = (10 marks)

6. (a) Protein: **[01 mark]**

(b) (i) Pepsin hydrolyse protein to peptides

(ii) Renin hydrolyse protein to peptides

(iii) Erepsin Peptidase hydrolyses peptide to Amino acid

(iv) Trypsin hydrolyses protein to peptides

Any two enzymes

@ 2 x 2 (04 marks)

(c) (i) Repair of damaged cells and tissues.

(ii) Growth and development of living things.

(iii) Formation of usefully chemicals such as hormones and enzymes.

(iv) Act as components of cell membrane e.g. Glycoprotein

(v) Transportation of oxygen by haemoglobin which is made upo of protein.

(vi) formation of bod organs

(vii) absorption of excess body fluids

(viii) used in energy production

@ 01 mark x 05 [05 marks]

7. (a) (i) First Aid Kit: Is the small box in which items used for first aid are kept. **02marks**

(b) (i) Painkillers: Reliving pain/reducing pain.

(ii) Bandages: Keeping dressing in place and scaring fractures or dislocated bores.

(iii) Cotton wool: Cleaning and drying wounds.

(iv) Liniment : Reducing muscle pain

@ 02 x 4 (08 marks)

8. (a)

	ORGANISM	KINGDOM	PHYLUM
(i)	Common mould	Fungi	Zygomycota
(ii)	Moss plant	Plantae	Bryophyta
(iii)	Paramecium	Protoctista	Ciliophora
(iv)	Mushroom	Fungi	Basidiomycota
(v)	Fern plant	Plantae	Filicinophyta
(vi)	Yeast	Fungi	Ascomycota

@ 01 x 06 (06 marks)

(b) (i) Fungi store food as glycogen while plantae store starch.

(ii) Fungi are heterotrophs (saprophytes) while plantae are autotrophs.

(iii) Fungi has cell wall made of Clotting while plantae have cellulose cell wall.

(iv) fungi has no chloroplast while Plantae has

Any two points **@ 02 x 2 (04 marks)**

9. (a) Guard cells: - To carry out gaseous exchange in plants./ control closing and opening of stomach **01mark**

(b) P-Nucleus: - Control all cellular activities

- Cetre hereditary information

- Control cell division

Q -Vacuole - Ford storage

- Secrete and excrete waste from the cell

R – Chloroplast - Site for photosynthesis

S – Cell wall: - Protect and support the cell

- Gives a definite shape
- Allows passage of water and minerals

Do ½ part

01 – Role

01 ½ x 4 (06 marks)

(c) (i) Enable plants to obtain carbondioxide as a raw material of photosynthesis.

(ii) Enable plants to obtain oxygen required to generate energy.

(iii) Enable plants to remove excess carbondioxide gas during the right.

(iv) enable plants to remove excess water

@ 01 x 3 (03 marks)

10. Respiration: Refers to the step by step breaking down of food to release energy.

Differences between Aerobic and anaerobic respiration.

(i) Aerobic respiration oxygen is used to generate energy while anaerobic oxygen is not used.
(ii) Aerobic produce large amount of energy while anaerobic produce little amount of energy.
(iii) Water is produced during aerobic respiration while anaerobic do not produce water.
(iv) Food substances are completely broken down during aerobic while during anaerobic food substances are broken down incompletely (partially)
(v) Aerobic start in the cytoplasm and ends in mitochondria while anaerobic take place in cytoplasm.
(vi) Aerobic respiration carbondioxide and water are end products while anaerobic lactic acid in animal and alcohol in plants are produces

Introduction – 02 marks
Main body 02 x 6 = 12 marks
Conclusion 01 mark

Wabillah Tawfiq